Electron-Ion Path Integral Monte Carlo with the Pair-Product Action

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(Dated: May 5, 2026)

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Keywords: Path Integral; Monte Carlo; Quantum Mixture; Bosons; Fermions; Two Component Plasma; Thermodynamics; Structure; Superfluidity; Sign Problem; Metallic Hydrogen; Molecular Hydrogen; Hydrogen Atom; Hydrogen Molecule; Pair Coulomb Density Matrix

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I. INTRODUCTION

In order to study the quantum statistical properties of the H fluid mixture through a computer experiment it is necessary to add to our simulation box of volume Ω an equal number $N/2$ of electrons (e) of mass m_e and of protons (p) of mass m_p and let them interact with the usual Coulomb pair potential $\pm 1/r$. We thus obtain a fluid of number density $n = N/\Omega$. We can then study the properties of H in thermal equilibrium at an inverse temperature $\beta = 1/k_B T$ with k_B Boltzmann constant and T the absolute temperature. If the laws of quantum statistical physics and of Coulomb are meaningful we should be able to reproduce the thermodynamic and structural properties of H that we observe in nature.

In our reduced units we will measure lengths in units of a Bohr radius $a_B = \hbar^2/m_e e^2$ and energies in units of two Rydbergs e^2/a_B, i.e. of a Hartree. The Hamiltonian operator and the density matrix in reduced units will then read

$$H = K + V = -\sum_{\alpha=1}^{N} \lambda_{i_\alpha} \nabla_\alpha^2 + \sum_{\alpha<\beta=1}^{N} \phi_{i_\alpha i_\beta}(r_{\alpha\beta}), \tag{1.1}$$

$$\rho = e^{-H/T}. \tag{1.2}$$

where we indicate with a Greek index the particle label and with a Roman index the species label so that for example i_α is the species label of particle α, either 1 for the electrons or 2 for the protons, $\boldsymbol{r}_\alpha$ is the position vector of particle α, and $r_{\alpha\beta} = |\boldsymbol{r}_{\alpha\beta}| = |\boldsymbol{r}_\alpha - \boldsymbol{r}_\beta|$ is the distance between particles α and β. The partial pair potential is the usual Coulomb potential $\phi_{ij} = \epsilon_{ij}/r$ with $\epsilon_{ii} = +1$ for all i and $\epsilon_{ij} = -1$ for $i \neq j$. $\lambda_i = \hbar^2/2\mu_i$ where the reduced masses are $\mu_1 = 1$ and $\mu_2 = m_p/m_e \approx 1836.15$. Temperatures are in units of ≈ 315775K.

The coordinate representation of the high temperature density matrix for distinguishable particles is [1]

$$\rho(R, R'; \tau) = \rho_0(R, R'; \tau) e^{-U(R,R';\tau)}, \tag{1.3}$$

$$\rho(R, R'; \tau) = \prod_{\alpha=1}^{N} (4\pi\lambda_{i_\alpha}\tau)^{-3/2} e^{-(\boldsymbol{r}_\alpha - \boldsymbol{r}'_\alpha)^2/4\lambda_{i_\alpha}\tau}, \tag{1.4}$$

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where $\tau = \beta/M$ for large M. So that the coordinate representation of the thermal density matrix for identical particles can be obtained from the following discretized path integral [2]

$$\rho(R, R'; \beta) = \sum_{\mathscr{P}} \mathrm{sgn}(\mathscr{P}) \int \rho(\mathscr{P}R, R_1; \tau)\rho(R_1, R_2; \tau) \cdots \rho(R_{M-1}, R'; \tau) \prod_{k=1}^{M-1} dR_k, \tag{1.5}$$

where $R_k = (\boldsymbol{r}_{1,k}, \boldsymbol{r}_{2,k}, \ldots, \boldsymbol{r}_{N,k})$ is the kth timeslice. Here $\boldsymbol{r}_{\alpha,k}$ is the position vector of particle α of species i_α at timeslice k in the path integral. The sum is over the permutations $\mathscr{P}$ of identical particles (of the same species) and $\mathrm{sgn}(\mathscr{P})$ is the sign of the given permutation

$$\mathrm{sgn}(\mathscr{P}) = \begin{cases} +1 & \text{Bose-Einstein statistics} \\ \begin{cases} +1 & \mathscr{P} \text{ an even permutation} \\ -1 & \mathscr{P} \text{ an odd permutation} \end{cases} & \text{Fermi-Dirac statistics} \end{cases} \tag{1.6}$$

While a Monte Carlo integration [3] of the path integral (1.5) (PIMC) can be carried out exactly for bosons, for fermions the alternating signs in the permutation sum becomes less and less efficient as N and/or M increase [4]. This is known as the 'fermions sign problem'. An approximate way to circumvent this problem is the Restricted Path Integral Monte Carlo (RPIMC) method of Ceperley [2, 4]. In our simulations we will always compare the PIMC for bosons and the RPIMC for fermions with a free fermions restriction.

For the inter-action U we will adopt the end-point approximation for the pair-product action [1]

$$U(R, R'; \tau) \approx \sum_{\alpha<\beta} P_{i_\alpha i_\beta}(\boldsymbol{r}_{\alpha\beta}, \boldsymbol{r}_{\alpha\beta}; \tau), \tag{1.7}$$

where for P_{ij} we choose the pair Coulomb density matrix of Pollock [5]. Namely

$$P_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau) = -\ln \left[\frac{\rho_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau)}{\rho^0_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau)} \right], \tag{1.8}$$

where

$$\frac{\partial \rho_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau)}{\partial \tau} = \left[\lambda_{ij} \boldsymbol{\nabla} - \frac{\epsilon_{ij}}{r} \right] \rho_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau). \tag{1.9}$$

Here $\lambda_{ij} = \lambda_i + \lambda_j$ and $\rho^0_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau) = (4\pi\lambda_{ij}\tau)^{-3/2} \exp[-(\boldsymbol{r} - \boldsymbol{r}')^2/4\lambda_{ij}\tau]$ satisfies the same Eq. (1.9) but for $\epsilon_{ij} = 0$ for all i, j.

TABLE I. Cases treated in our simulations for the H fluid mixture. We show the total number of particles N, the reduced number density $n = N/\Omega$, the reduced temperature T, the total kinetic energy per particle $e_K = \langle K \rangle/N$, the total potential energy per particle $e_V = \langle V \rangle/N$, the superfluid fraction for species i, $f_s(i)$. In the statistics column "b" stands for bosons, "f" for fermions. The simulation lasted more than 10^8 Monte Carlo steps.

case	statistics	T	M	n	N	Ne_K	$-Ne_V$	$f_s(1)$	$f_s(2)$
A	b (PIMC)	0.15	20	0.3	30	...	...	...	...
B	f (RPIMC)	0.15	20	0.3	20	...	...	...	...
A	b (PIMC)	0.15	20	0.16	30	...	...	...	...
B	f (RPIMC)	0.15	20	0.16	20	...	...	...	...

Appendix A: Description of our PIMC and RPIMC algorithms

In the RPIMC we reject the move whenever the product $D^0_1 D^0_2$ changes sign, where

$$D^0_i = \det \left|\left| (4\pi\lambda_i\tau)^{-3/2} e^{-(\boldsymbol{r}_\alpha - \boldsymbol{r}_\beta)^2/4\lambda_i\tau} \right|\right|_{\alpha\beta}, \tag{A1}$$

is the order $N/2$ determinant such that $i_\alpha = i_\beta = i$.

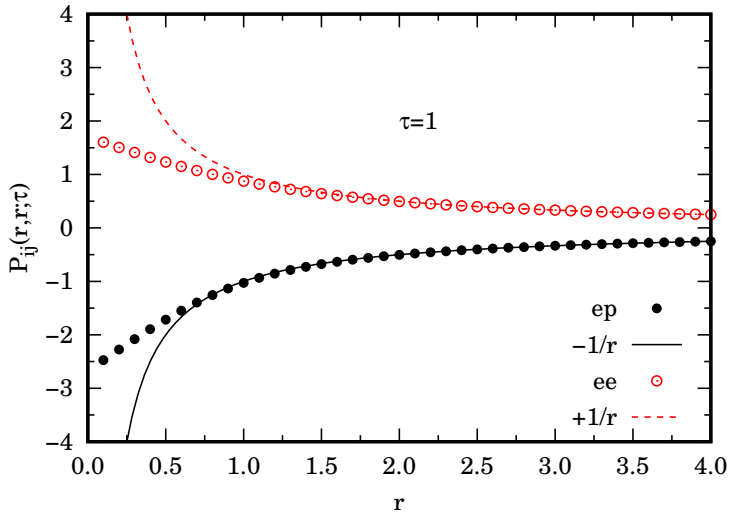

FIG. 1. We show the *ee* and *ep* pair Coulomb density matrix of Eq. (1.8) for $\tau = 1$. We compare it with the bare Coulomb potential. From the figure we can see that the cusp condition [5] $P_{ij}(\boldsymbol{r}, \boldsymbol{r}; \tau) \sim r\epsilon_{ij}/2\lambda_{ij}$ is satisfied at small r. The distance r is in units of a Bohr radius.

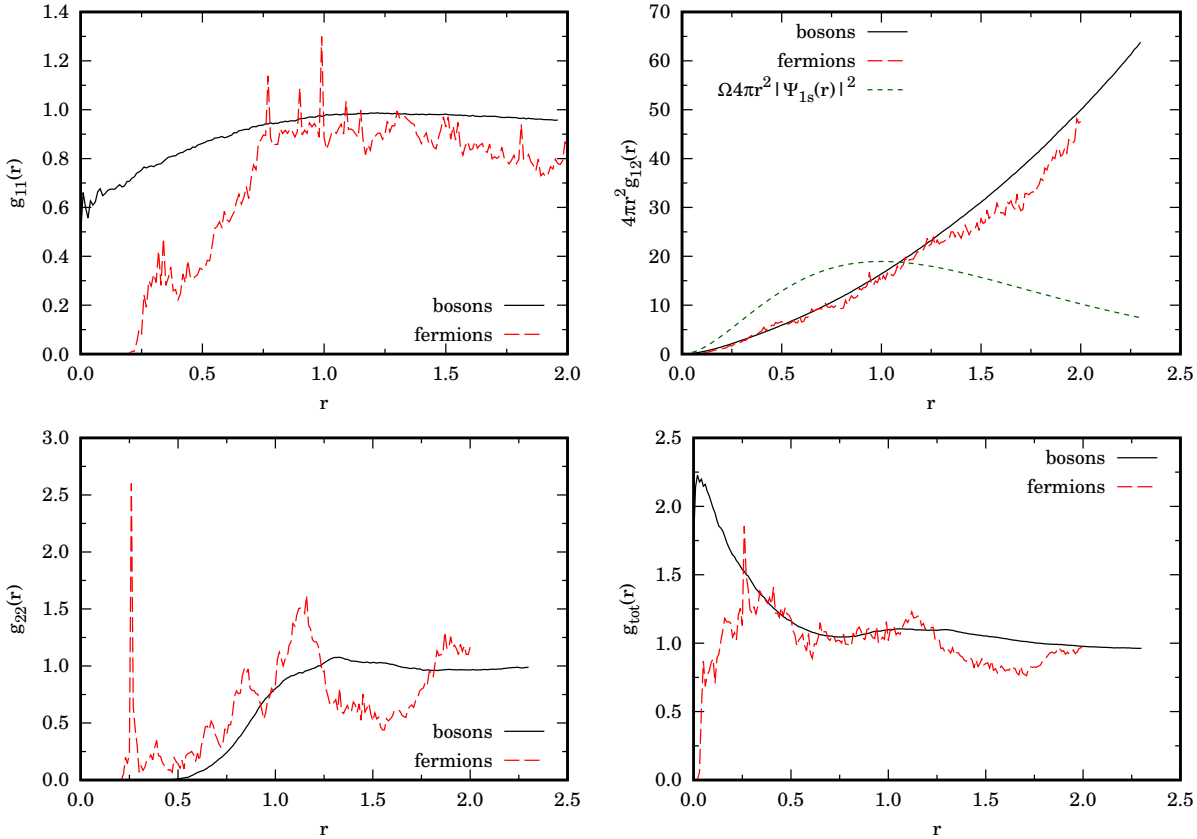

FIG. 2. We show the partial and total RDF of the H fluid for cases A and B in Table I. The distance r is in units of a Bohr radius.

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Funding

None declared.

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